

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

Date : 13/01/2026

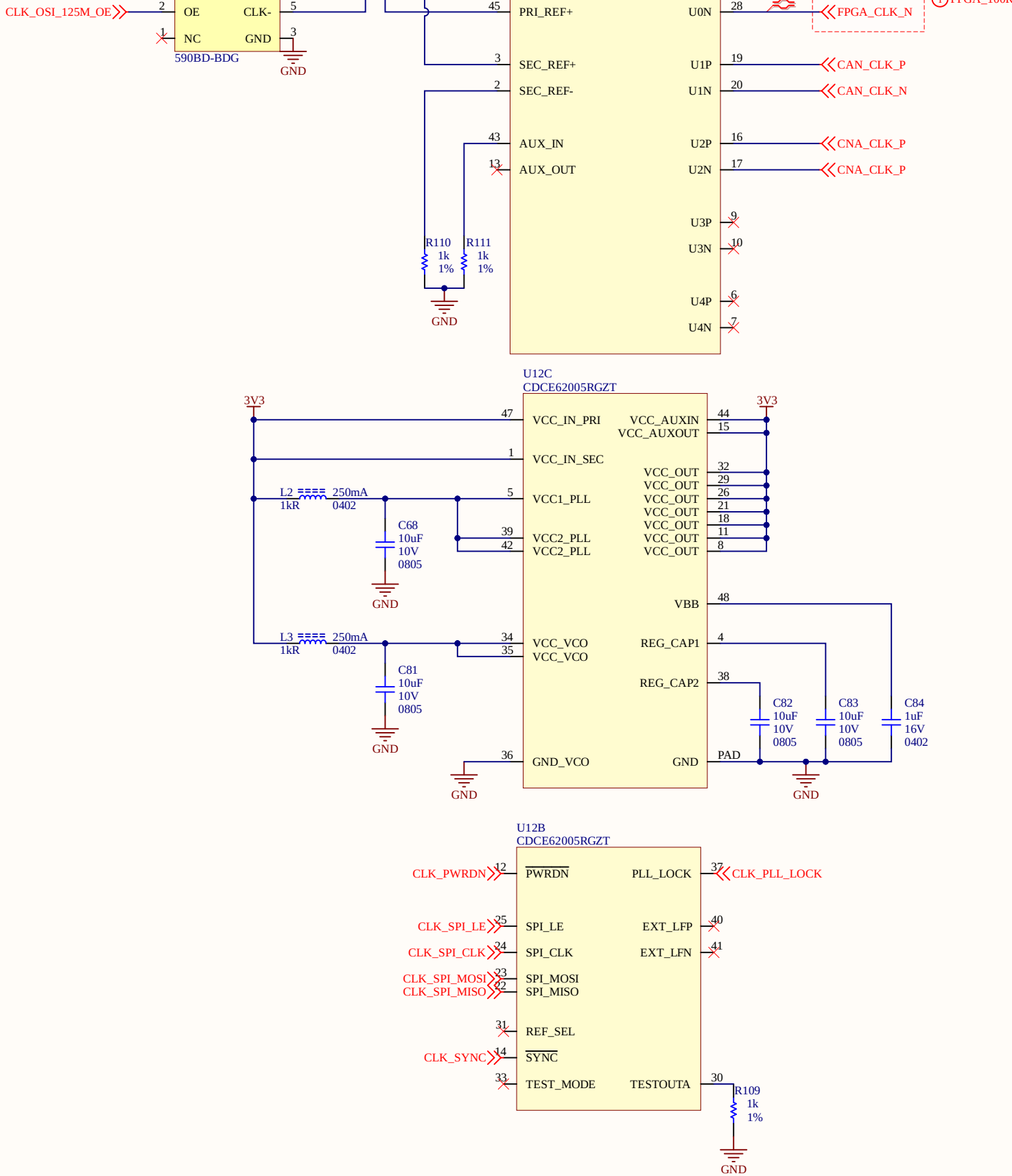
Revision history	

Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512



Project Title		
S7CAS-PLEIADES-CARTE-MERE		
Global Project		
Pleiades		
Size	Group	Revision
11x17	01	1.00
Date	Sheet	
13/01/2026	1 of 16	
Filename		Designers
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

CLOCK



CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf
CLOCK BUFFER - CDCE62005.pdf

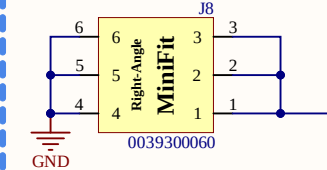
-Quand tu es en LVDS faut que branche le SEC à une résistance (CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf)

Sheet Name			CLOCK		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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11x17	01			1.00	
Date	13/01/2026		Sheet	2	of 16
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

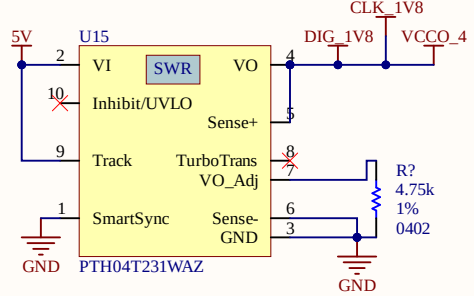
ALIMENTATION

24V

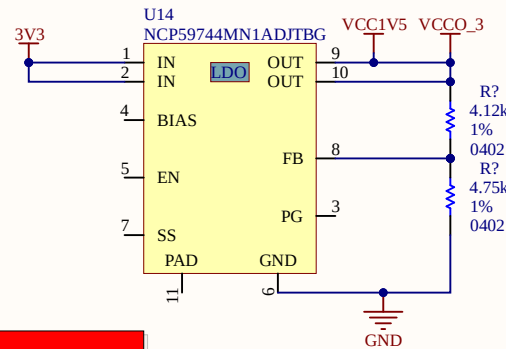
METTRE +24V APRÈS LES PROTECTIONS



1V8

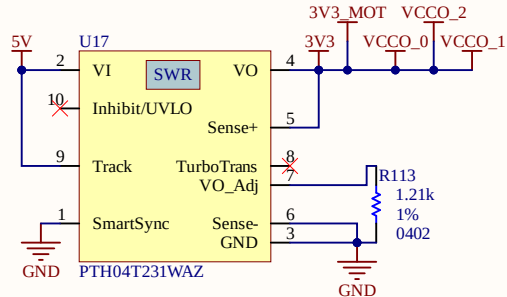


1V5

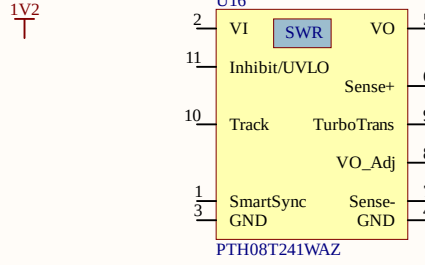


Soft Start?

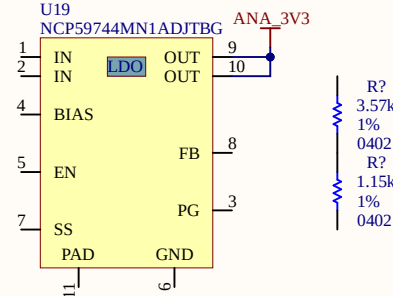
3V3



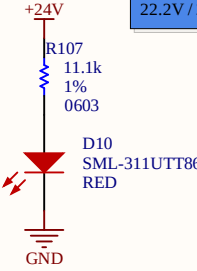
1V2



ANA_3V3

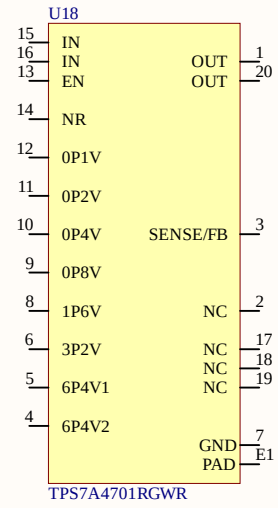


LEDS 24V



24V - 1.8V = 22.2V
22.2V / 20mA = 11100 Ohms

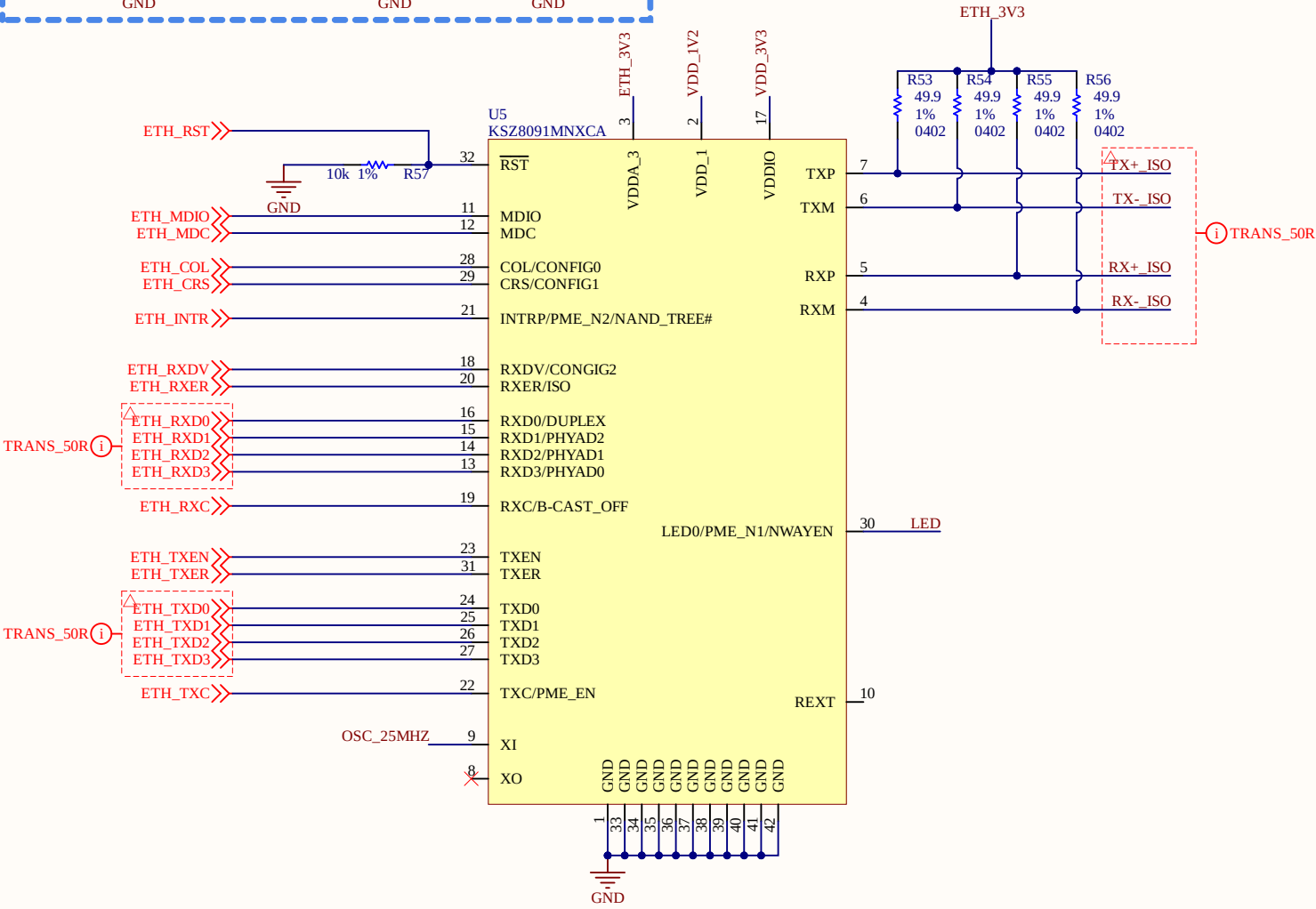
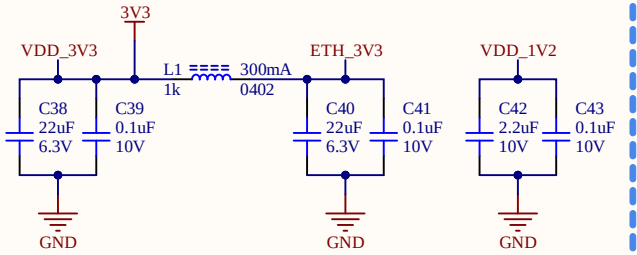
VCCO_3 à 1.5V pour le DDR3
VCCO_4 à 1V8 pour le control du DAC et ADC
Ne pas oublier les BCOOD et EN



Sheet Name			ALIM		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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S7CAS-PLEIADES-CARTE-MERE_P01_ALIM.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

ETHERNET
ALIMENTATION

ETHERNET - PHY - KSZ8091MNXCA.pdf

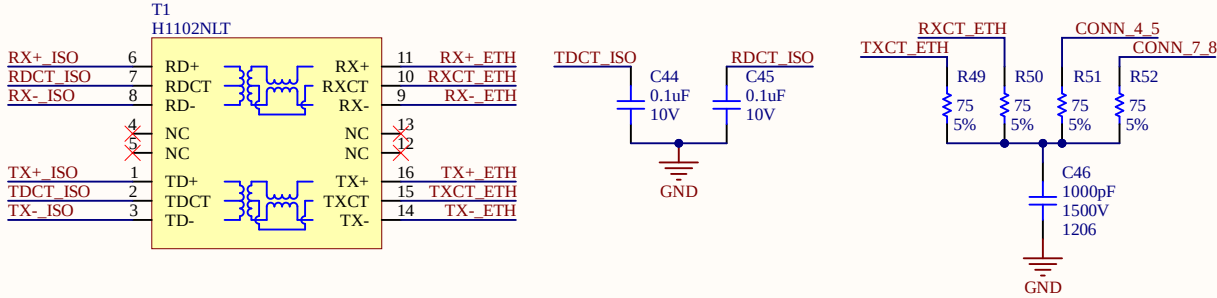


TRANS_50R

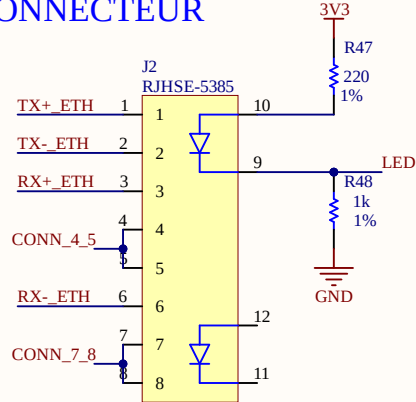
TRANS_50R

-Isolation du ETHERNET 1.5kV
-Ajout d'un condensateur de 1.5kV
-Ligne de transmission de 50R

ISO

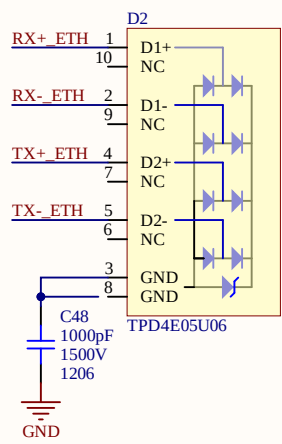


CONNECTEUR

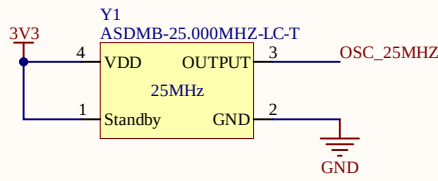


Pin #	Wire Color Legend	Signal
1	White/Green	TX+
2	Green	TX-
3	White/Orange	RX+
4	Blue	TRD2+
5	White/Blue	TRD2-
6	Orange	RX-
7	White/Brown	TRS3+
8	Brown	TRD3-

ESD

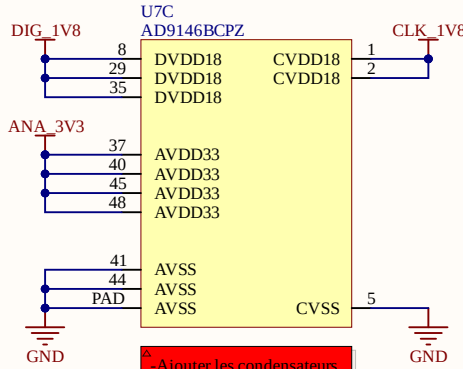
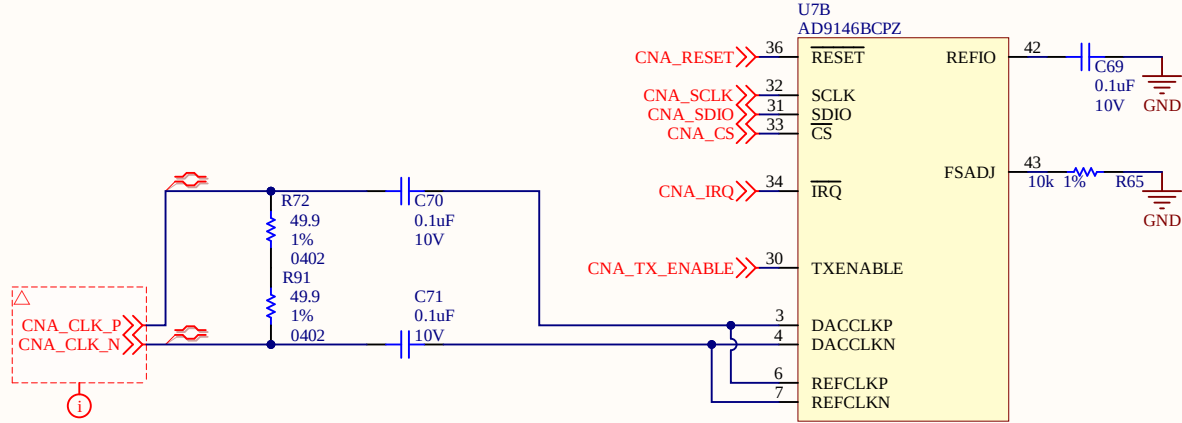
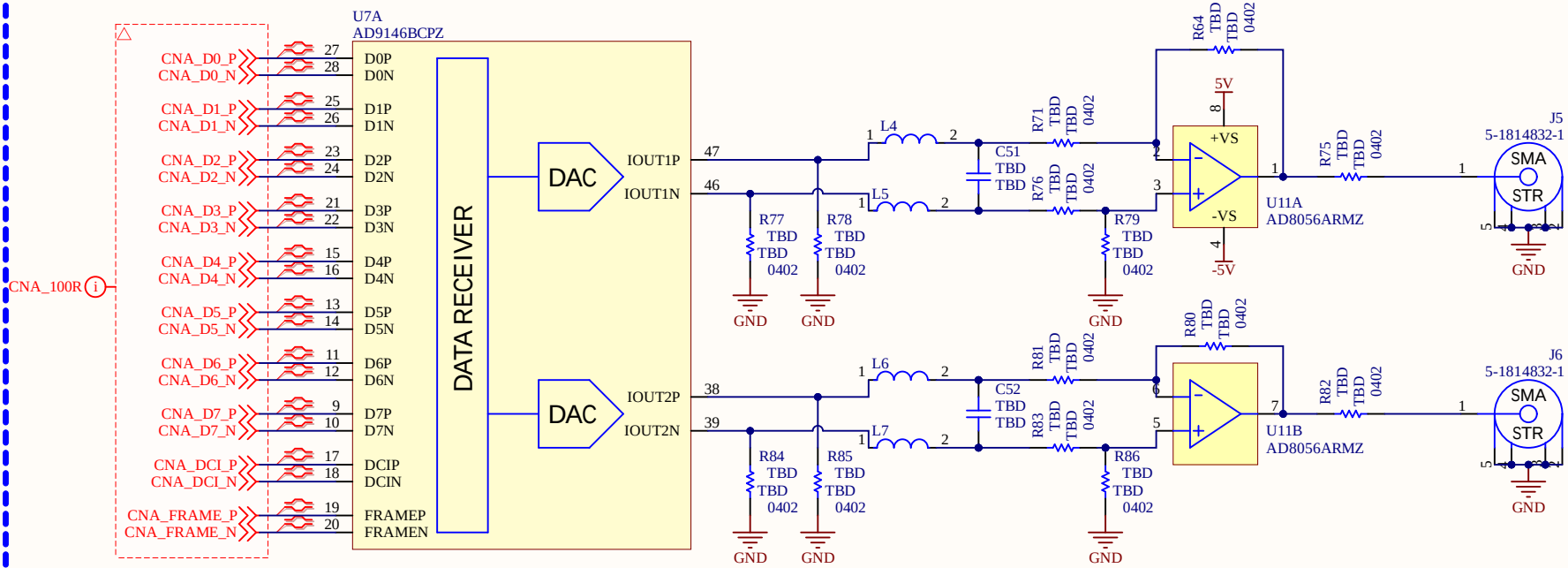


OSCILLI



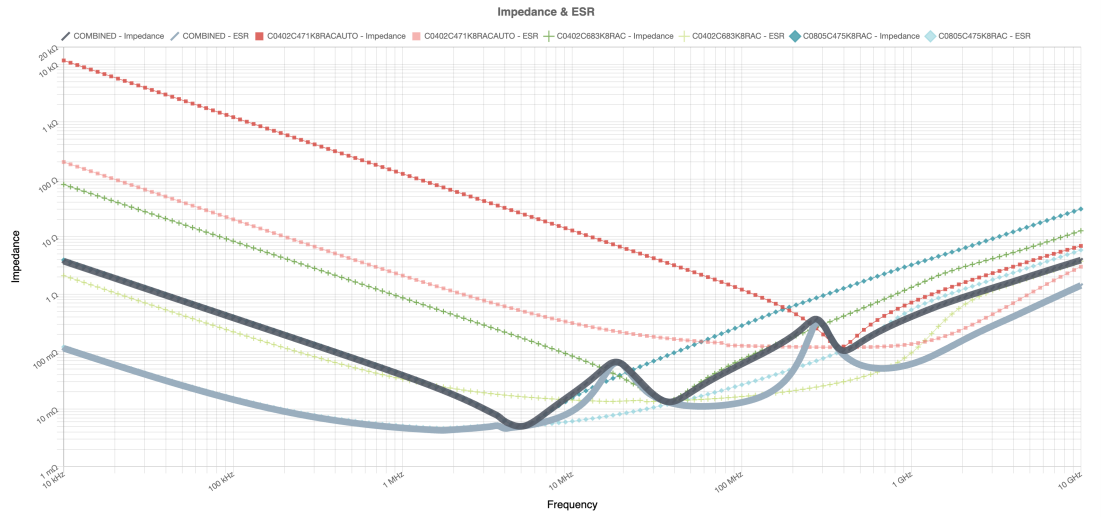
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Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	4 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

CNA (DAC)

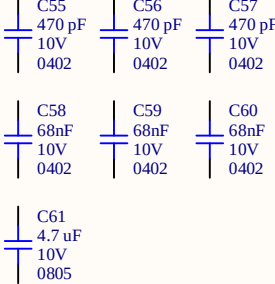


-Ajouter les condensateurs

-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4



Legend	Tolerance	Part Number	Cap.	V _{DC} *	Dielectric	Qty.	Bias (V)	Amb. (°C)	Add	Remove	
-			Combined	4.3 uF	10	X7R	1	0	25		X
■	10%	C0402C471K8RACAUTO	470 pF	10	X7R	3	0	25	+	X	
+	10%	C0402C683K8RAC	68 nF	10	X7R	3	0	25	+	X	
◆	10%	C0805C475K8RAC	4.7 uF	10	X7R	1	0	25	+	X	



Sheet Name		CNA	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
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11x17	01	1.00	
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S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

CAN (ADC)
DECOUPLAGE

OPAMP - ADA4927-1_ADA4927-2.pdf
P.20

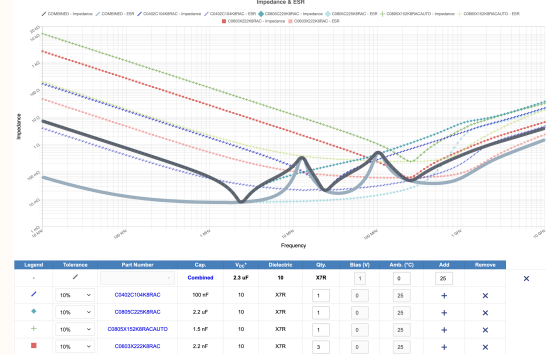
Ajuster les valeurs des résistances

Ajuster les valeurs des résistances

ADC - ADS62P48.pdf
Page 63

Ajouter les condensateurs

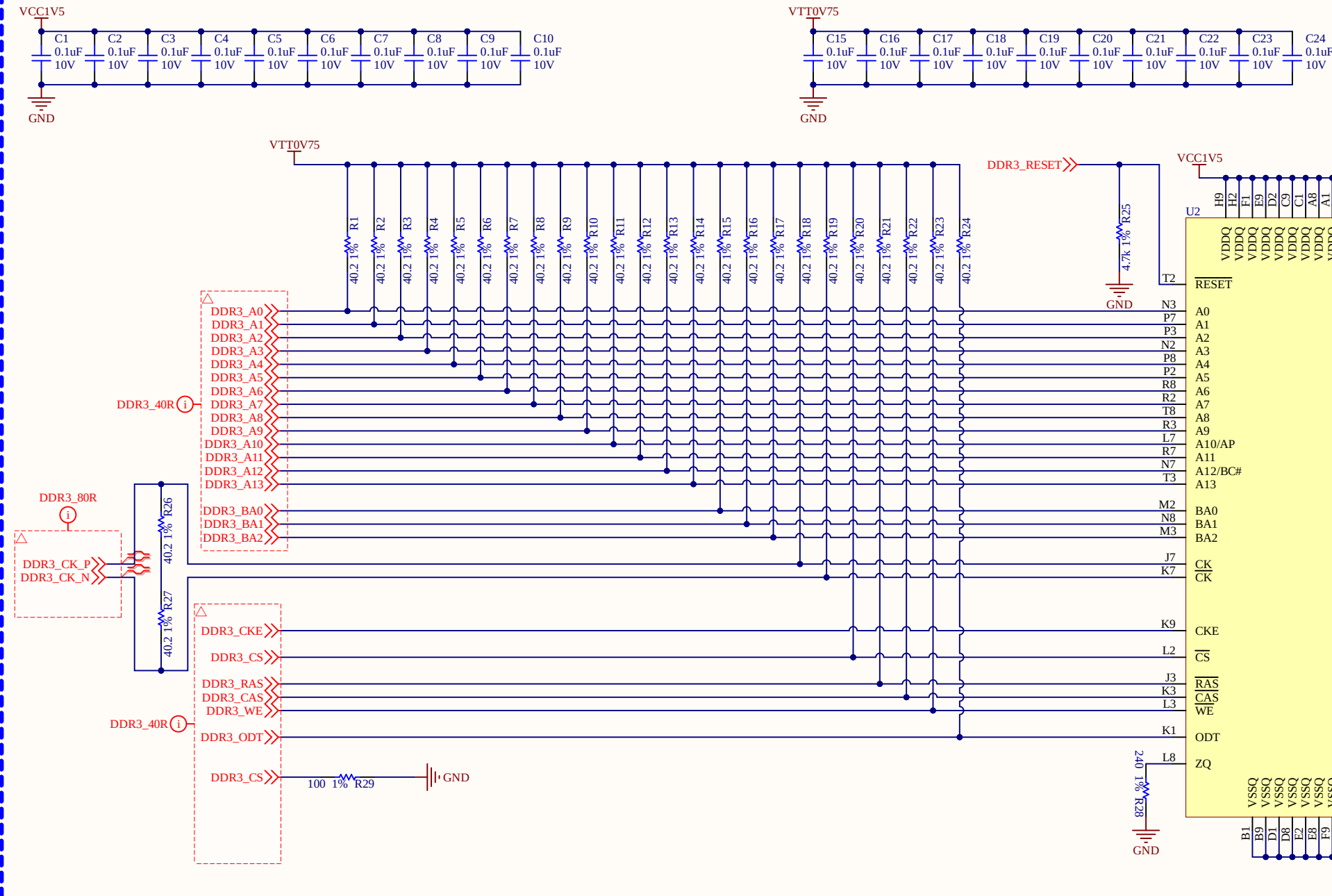
-Choix de OP-AMP pour isolé et pour passer d'analog a digital
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Nous avons pris uniquement AMPLI-OP puisque transfo bloque le DC et on a une composante DC



C62	100 nF	C63	2.2 uF	C64	1.5 nF	C65	2.2 nF	C66	2.2 nF	C67	2.2 nF
10V	0402	10V	0805	10V	0805	10V	0603	10V	0603	10V	0603

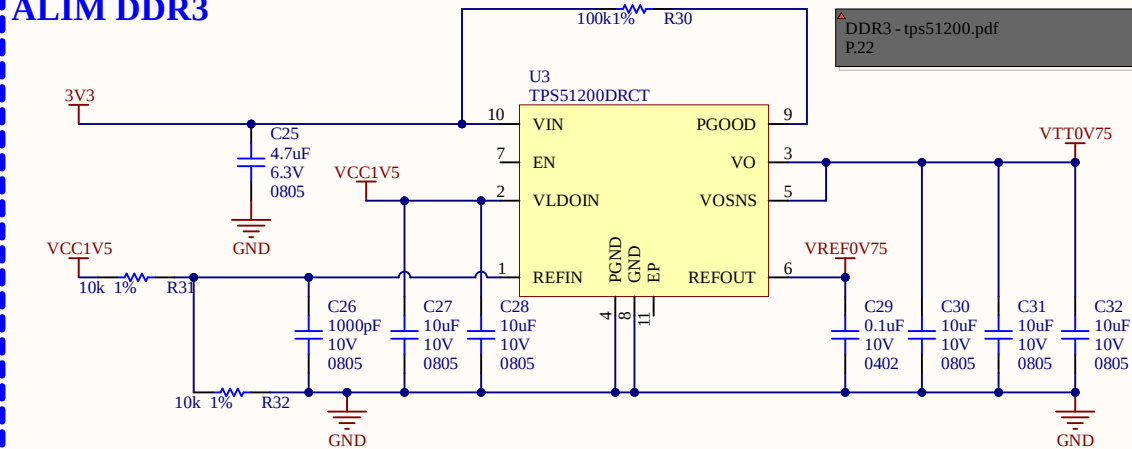
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Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	11x17	Group	01		Revision
Date			13/01/2026		1.00
Date			Sheet	6	of 16
Filename			S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc		Designers
					Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

FPGA



ANALYSER LES COURANTS POUR LE VTT
AJUSTER LES CONDENSATEURS

ALIM DDR3



-Ligne de transmission de 40R
-Resistance parallele sur la ligne de transmission de 40.2Ohms
-Clock en pair différentiel 80R

Sortie PGOOD?
EN?

ACT	33.4 mW
RD	21.0 mW
WR	23.4 mW
READ I/O	32.7 mW
Write ODT	136.1 mW
Total RD/WR/Term Power	213.2 mW
ACT_STBY	14.9 mW
PRE_STBY	3.1 mW
ACT_PDN	2.9 mW
PRE_PDN	0.7 mW
REF	4.2 mW
Total Background Power	25.8 mW
Total DDR3 SDRAM Power	272.3 mW
TERM 2nd rank	0.0 mW

Sheet Name

DDR3

Project Title

S7CAS-PLEIADES-CARTE-MERE

Global Project

Pleiades

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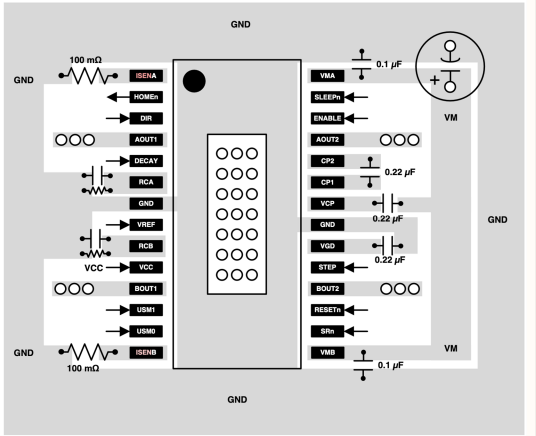
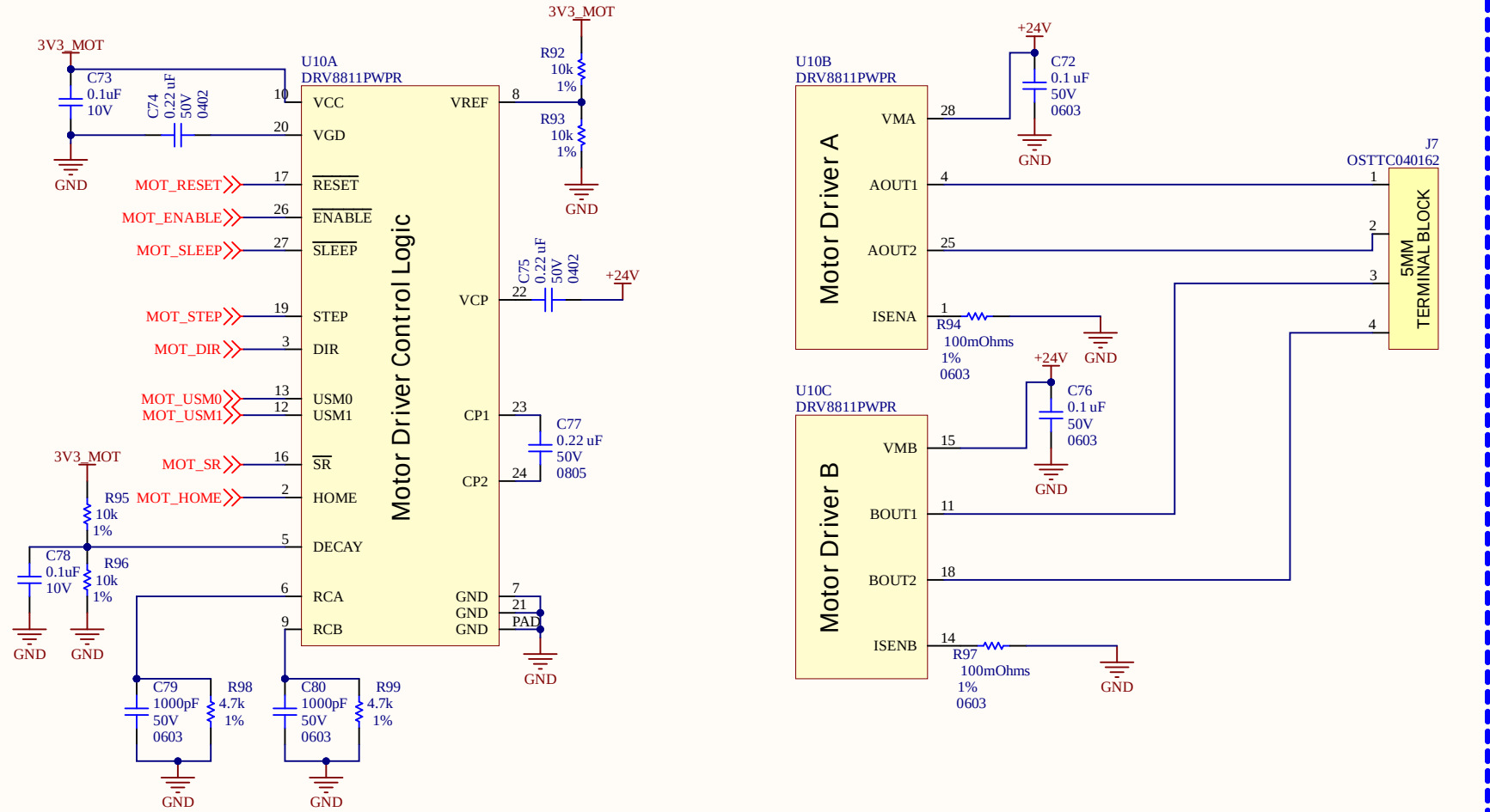
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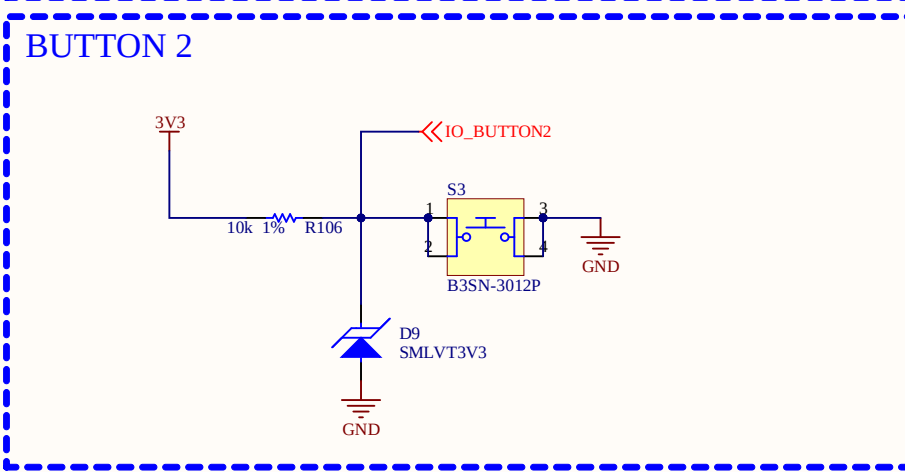
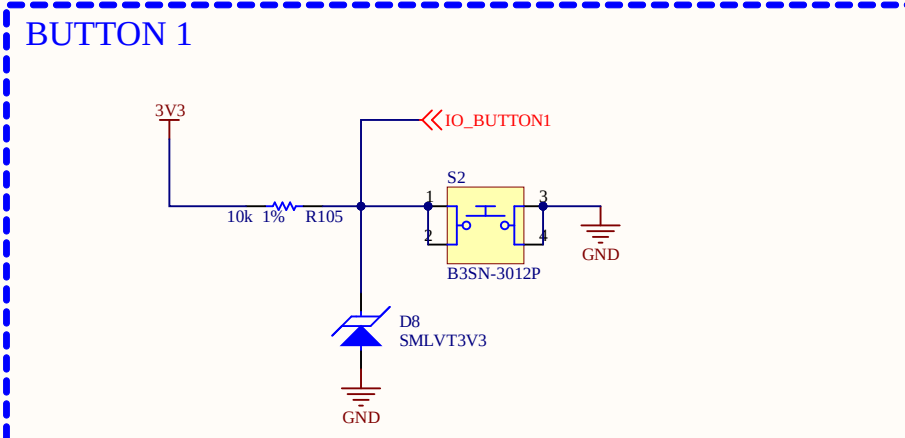
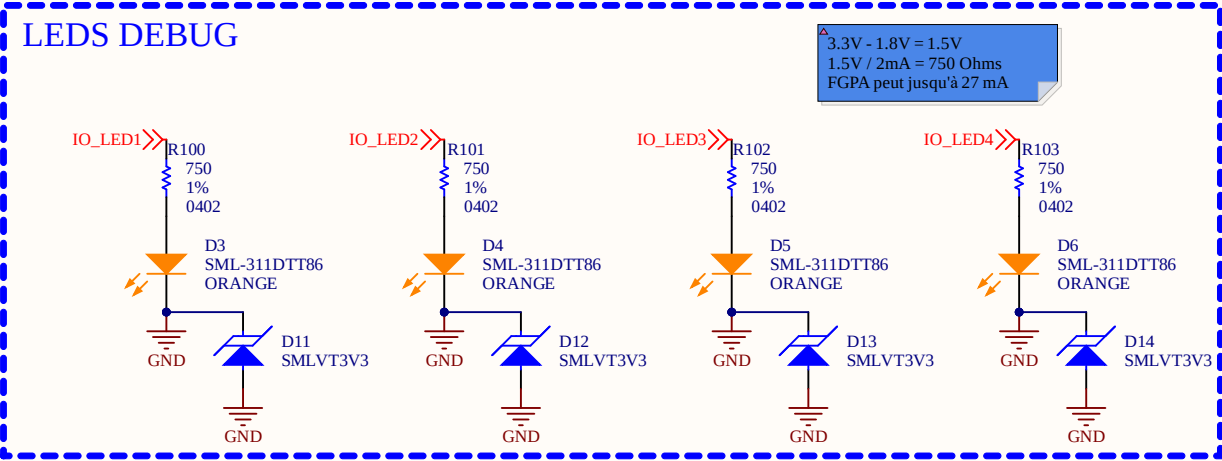
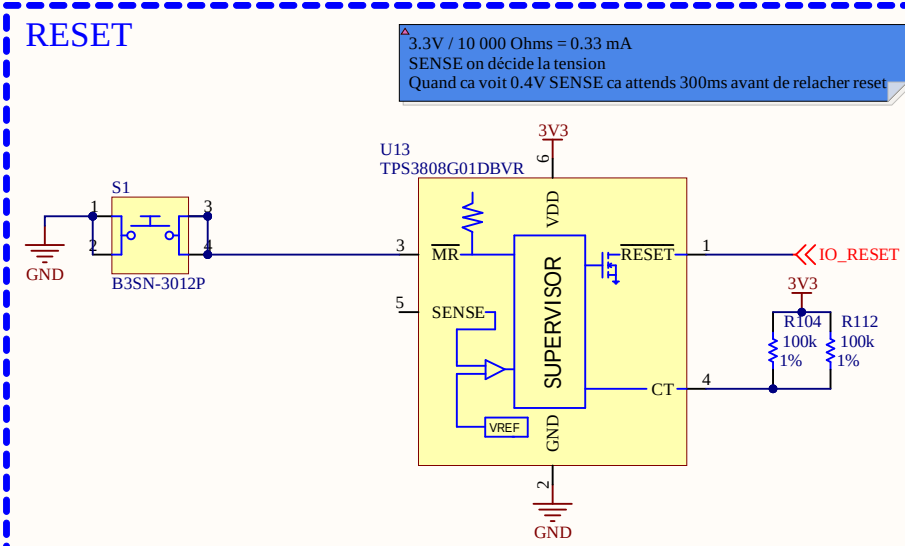
Designers

Mathieu Desautels (DESM1210)
Mathys Plante (PLAM1001)
Jonathan Bouchard

MOTOR

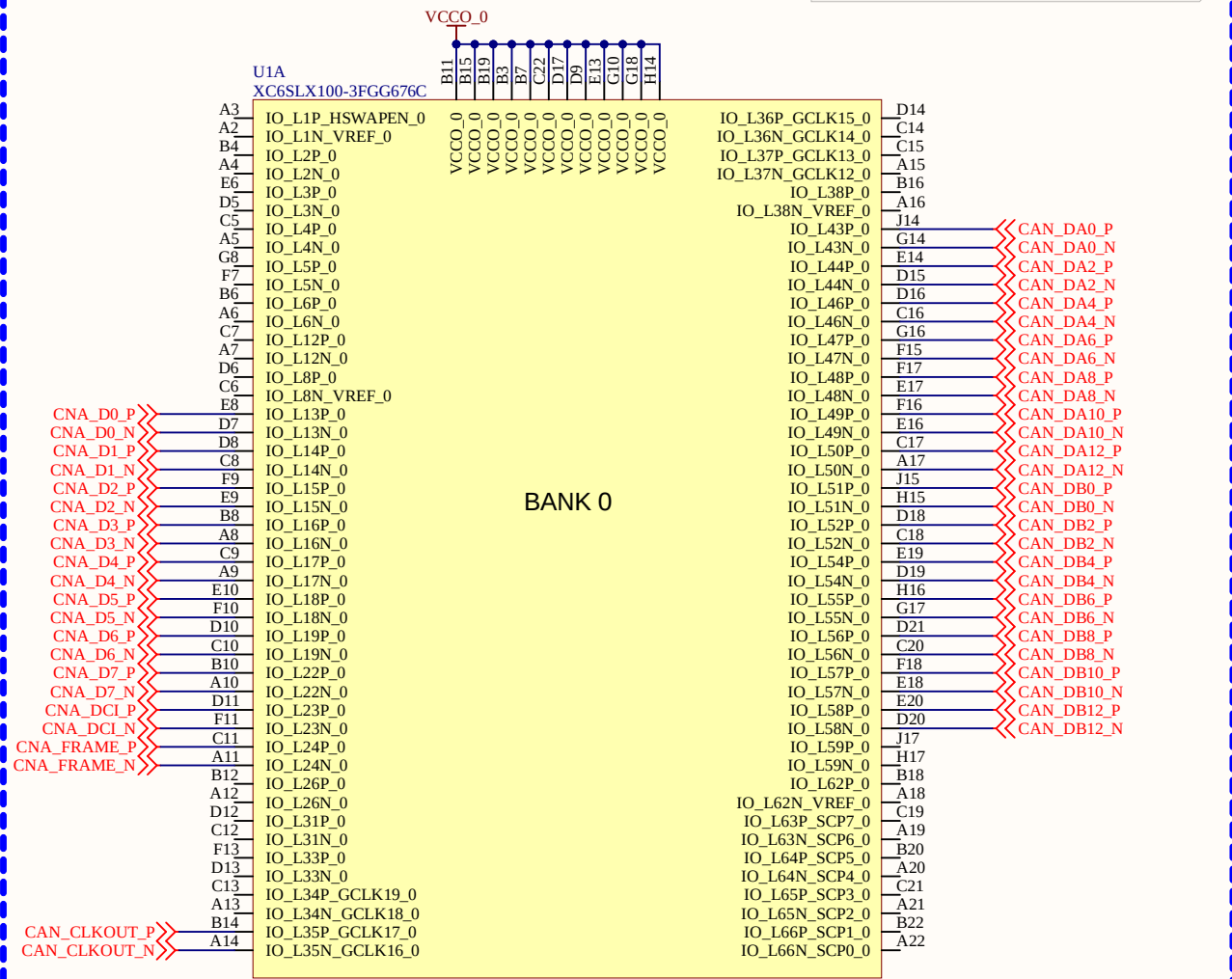


Sheet Name		MOTORS	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
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11x17	01	1.00	
Date	13/01/2026	Sheet	8 of 16
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_MOTORS.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



Sheet Name			IO		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_IO.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

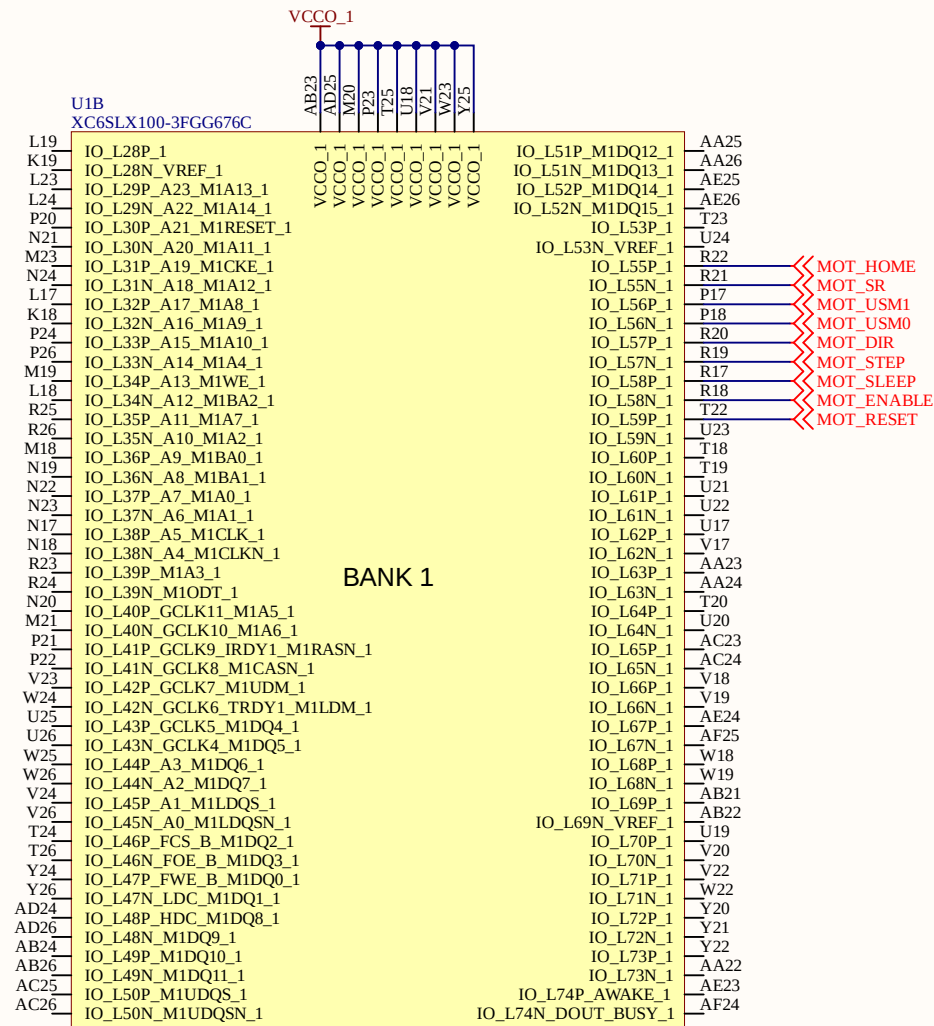


IO CAN est LVDS donc il faut DIFF_TERM = "TRUE"

- LVDS seulement sur BANK0 et BANK2
- Toute les CAN et CNA sont mit sur cette BANK

Sheet Name			FPGA - BANK 0		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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Size	Group			Revision	
11x17	01			1.00	
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Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

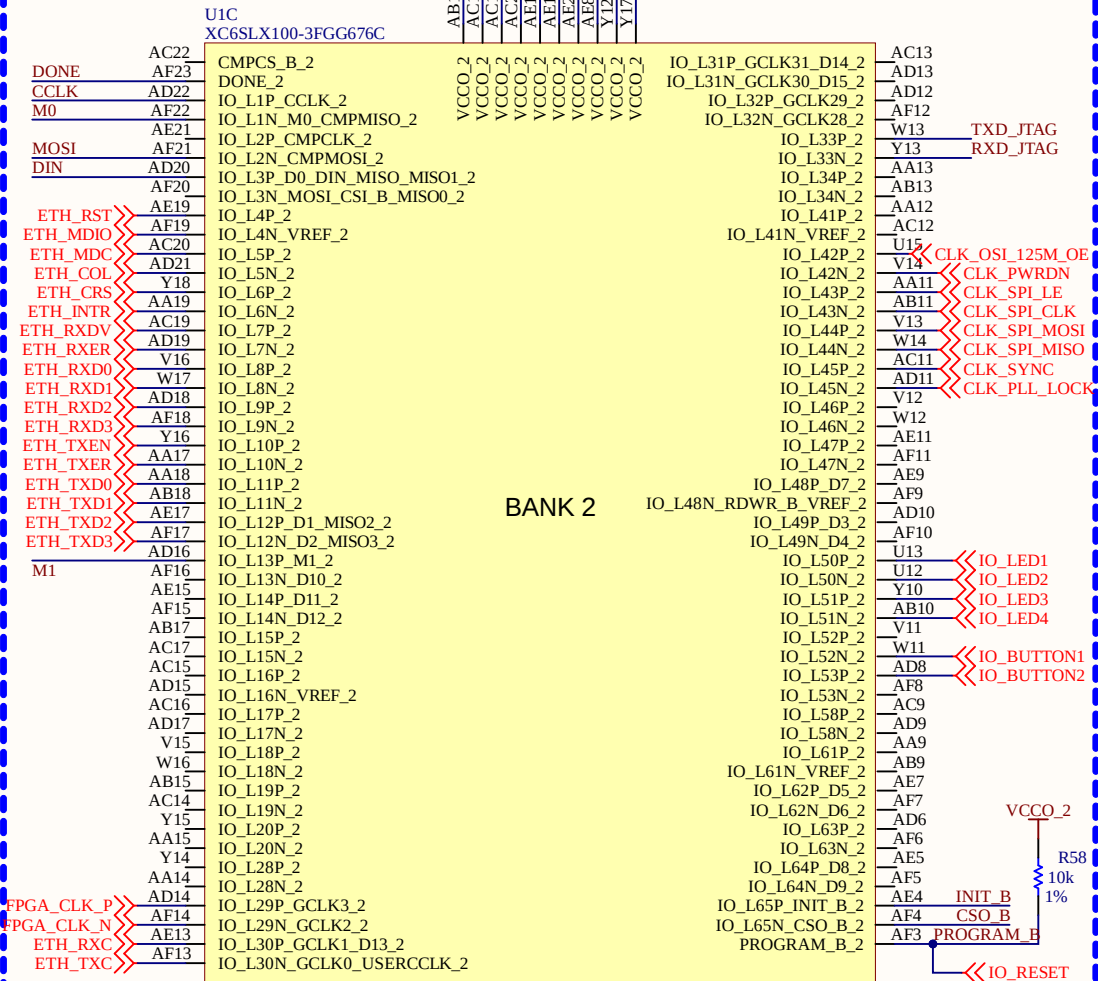
FPGA



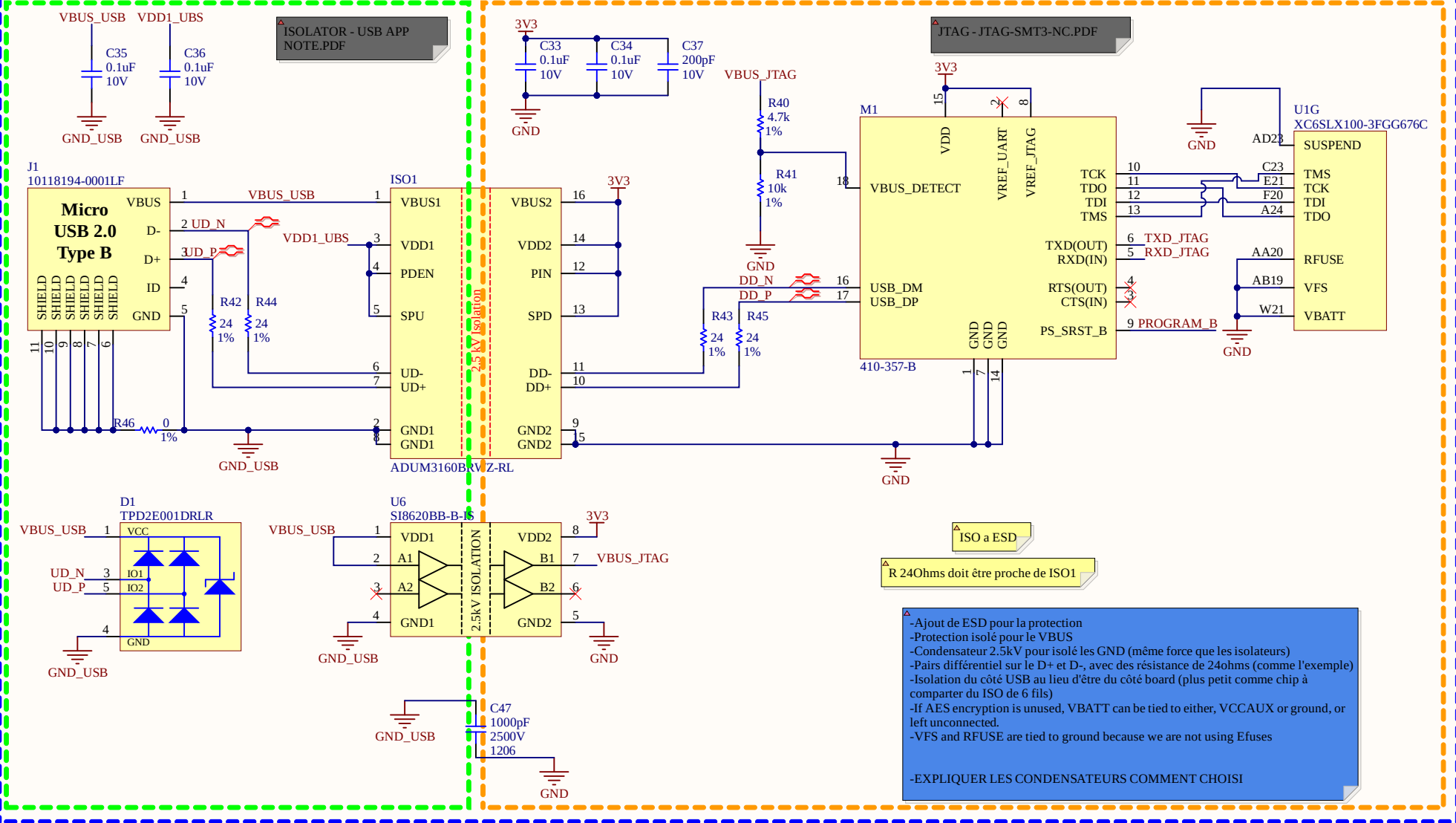
- Toute les controles de moteur sont sur cette BANK

Sheet Name			FPGA - BANK 1		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
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13/01/2026	11	16			
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK1.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

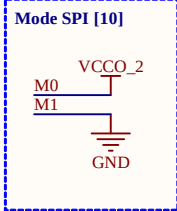
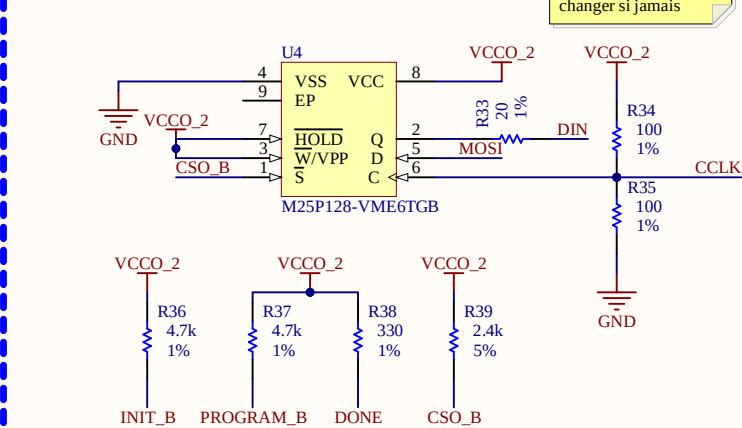
FPGA



USB



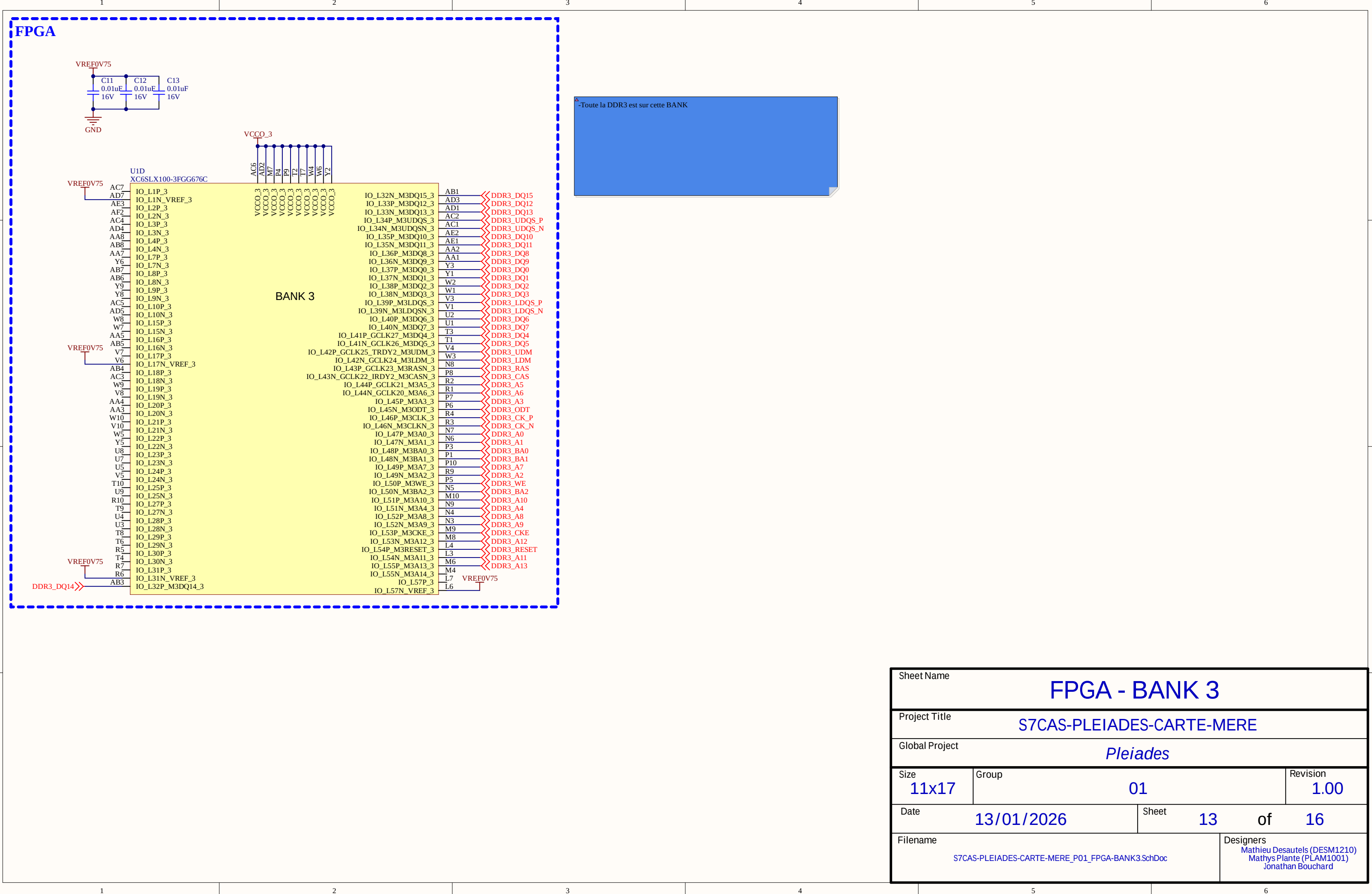
FLASH



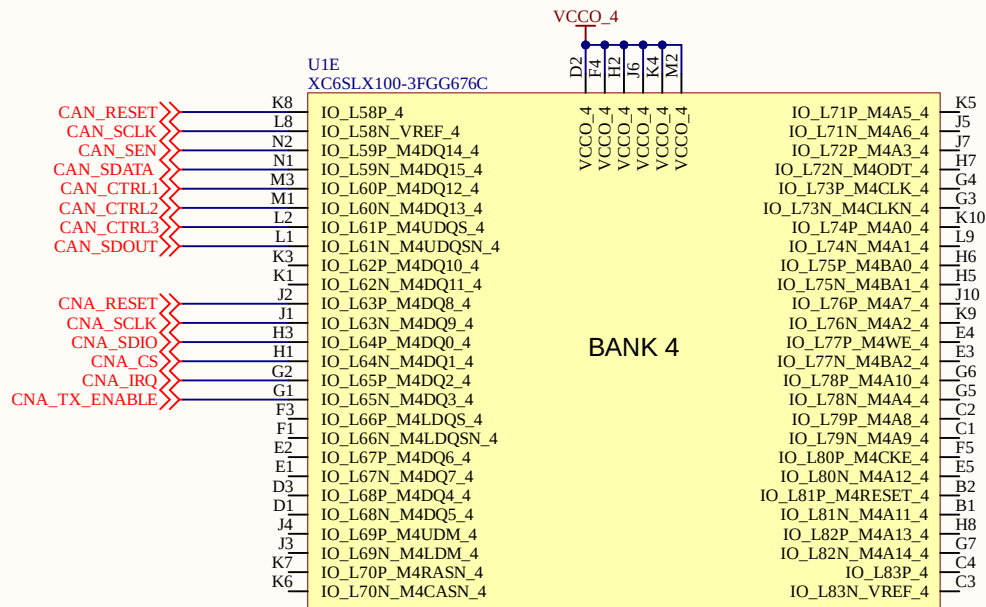
-M0 et M1 choisi pour avoir le mode SPI
- Ajout d'une résistance de 20 ohms sur Q
- Toute la logique de code est sur la BANK ainsi que toute la programmation

-Ajuster les condensateurs
-RESET ça va se placer où (question Mathys)

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLIADDES-CARTE-MERE		
Global Project			Pleiades		
Size	11x17	Group	01		Revision
Date			13/01/2026		Sheet
Filename			S7CAS-PLIADDES-CARTE-MERE_P01_FPGA-BANK2.SchDoc		Designers
					Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard



FPGA



- Toute le controle du CNA et CAN sont sur cette BANK
- Leur tension n'est pas la même que celle de leur données

Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group	Revision			
11x17	01	1.00			
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

